Network Functional Varying Coefficient Model

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Abstract

We consider functional responses with network dependence observed for each individual at irregular time points. To model both the inter-individual dependence as well as within-individual dynamic correlation, we propose a network functional varying coefficient (NFVC) model. The response of each individual is characterized by a linear combination of responses from its connected nodes and its own exogenous covariates. All the model coefficients are allowed to be time dependent. The NFVC model adds to the richness of both the classical network autoregression model and the functional regression models. To overcome the complexity caused by the network inter-dependence, we devise a special nonparametric least squares type estimator, which is feasible when the responses are observed at irregular time points for different individuals. The estimator takes advantage of the sparsity of the network structure to reduce the computational burden. To further conduct the functional principal component analysis, a novel within-individual covariance function estimation method is proposed and studied. Theoretical properties of our estimators are analyzed, which involve techniques related to empirical processes, nonparametrics, functional data analysis and various concentration inequalities. We analyze a social network data to illustrate the powerfulness of the proposed procedure.

KEY WORDS: Functional regression model; Kernel estimator; Network autoregression model; Nonparametric estimation.

1 Introduction

Functional linear regression is an important statistical tool for exploring regression relationships of data in the form of functions. Since introduced by Ramsay and Dalzell (1991), it is further studied by many researchers (Faraway, 1997; Chiou et al., 2003). In practice, the model has been successfully applied to a wide range of fields, which include gene expression analysis (Leng and Müller, 2006), image data modelling (Zhu et al., 2012), longitudinal data analysis (Müller, 2005; Huang et al., 2014) and many others.

The original form of functional linear regression model assumes a linear relation between a functional response and a number of exogenous covariates. Since the model has been proposed, it has been extended in various ways. For instance, Yao et al. (2005c) and Ferraty et al. (2012) incorporate functional covariates and investigate corresponding asymptotic properties. Hall and Hosseini-Nasab (2006) study the properties of functional principal component analysis. The additive structure and mixture components are considered by Müller and Yao (2008) and Chiou (2012) respectively. Zhu et al. (2012) further models multivariate functional responses by using a varying coefficient model. See Ramsay and Silverman (2007); Wang et al. (2016) for a comprehensive summary.

So far, most existing works assume that the functional curves are independently sampled. As a consequence, cross-curve dependence structure is not incorporated into the modelling framework. In this work, we take into account an important dependence structure, i.e., the network dependence, among the functional curves. This is timely and critically important as an increasing number of network data emerges in numerous applications (Zhao et al., 2012; Chen et al., 2013; Sojourner, 2013; Härdle et al., 2016;

Zou et al., 2017).

To incorporate the network dependence structure into the functional data modelling, we employ the idea of network autoregression model. Consider a network with N nodes, indexed with $i = 1, \cdots, N$. For the ith node, the response is recorded as $Y_i(t)$ and the associated covariate vector is $\boldsymbol{X}_i$. The network nodes are linked by the network relationships, which is recorded by an adjacency matrix $A = (a_{ij}) \in \mathbb{R}^{N \times N}$. Specifically, let $a_{ij} = 1$ if there is a link from the ith node to the jth node, otherwise $a_{ij} = 0$. We model $Y_i(t)$ by a linear combination of several factors. The first factor is $\boldsymbol{X}_i$, which is consistent with the classical functional linear regression models. The second factor is the overall response from the node's connected friends, i.e., $\sum_j a_{ij} Y_j(t)$, which is referred to as network autoregression term and quantifies the network influence. We would like to remark that this term brings much challenge to model estimation, since the responses of the focal node and its connected friends may not be observed at the same time point t. As a result, conventional estimation methods do not apply and novel estimators needed to be devised. In addition to these two important factors, to handle the special feature of functional data, we also take into account the within-curve dependence through using functional principal components.

The classic network autoregression model has been applied in the network regression literature. For instance, in social network analysis, researchers find that a user's posting behaviour is positively related to his/her following friends' activities (Zhu et al., 2017, 2020; Huang et al., 2020). In financial risk management, there is evidence showing that the market disturbance spreads through stock networks (Acemoglu et al., 2015; Zou et al., 2017; Zhu et al., 2019). However, several important limitations are worth pointing out. First, the model coefficients are assumed fixed across time. Therefore, the model cannot reflect time varying regression patterns. Second, the model assumes

the response $Y_i(t)$ to be observed across all individuals at the same time point t. As a result, it does not allow for responses observed at irregular time points. Finally, the model does not take into account of potential within-curve time dependence for $Y_i(t)$.

To this end, we propose a network functional varying coefficient (NFVC) model, which simultaneously incorporates time varying coefficients, irregular observation time points and network dependence structure. To perform estimation, a novel nonparametric conditional least squares estimation (LSE) approach is studied, which is related to but very different from the LSE for cross-sectional network autoregression model (Zhu et al., 2019). The estimator is capable of reducing the computational burden especially for sparse networks. Subsequently, given the estimated model, the functional principal component analysis (FPCA) is conducted to further account for the within-individual dependence. To this end, we devise a novel nonparametric method for estimating the covariance function under the network dependence structure, which is crucial for the FPCA procedure. The asymptotic properties are investigated for both parameter function estimation as well as FPCA results, which prompts us to assemble tools related to empirical process, nonparametric analysis, and various concentration inequalities on martingale sequences. Simulation studies demonstrated the satisfactory finite sample performance of our procedure. Lastly, we illustrate the proposed method by a real data example.

The rest of the article is organized as follows. We first describe the newly proposed model and define some notations in Section 2. We describe the parameter estimation procedure and the related theoretical results in Section 3. The estimation and the theory regarding the covariance function and functional principal component analysis related to the within node functional innovation process are studied in Section 4. The simulation studies are illustrated in Section 5 and we present a real data example in

Section 6. We conclude the paper in Section 7. All the detailed theoretical proofs are in the Supplementary Document.

2 Model and Notations

2.1 Main model

We consider a network with N nodes in the time period $0 \leqslant t \leqslant 1$. For each node i, we assume the observations are sampled from a functional trajectory $Y_i(t)$. To describe the link between different nodes, we use an adjacency matrix $\mathbf{A} = (a_{ij}) \in \mathbb{R}^{N \times N}$, where $a_{ij} = 1$ if the ith node follows the jth node, otherwise $a_{ij} = 0$. Following the convention, we let $a_{ii} = 0$. To capture the network dependence, we propose a network functional varying coefficient (NFVC) model

$$Y_i(t) = \boldsymbol{X}_i^\top \boldsymbol{\beta}(t) + \rho(t) m_i^{-1} \sum_j a_{ij} Y_j(t) + \eta_i(t) + \varepsilon_i(t). \tag{2.1}$$

In (2.1), the first term $\boldsymbol{X}_i^\top \boldsymbol{\beta}(t)$ describes the *nodal effect* of its own features $\boldsymbol{X}_i$. Specifically, $\boldsymbol{X}_i \in \mathbb{R}^q$ is an exogenous covariate vector and the associated regression parameter function $\boldsymbol{\beta}(t) \in \mathbb{R}^q$ captures time varying effects. The second term is the *network effect*, which reflects the influences of the connected nodes at the same time point. Here $m_i = \sum_{j=1}^N a_{ij}$, reflecting the out-degree of node i. Hence $m_i^{-1} \sum_j a_{ij} Y_j(t)$ characterizes the average responses of node i's connected friends. The network connection strength is quantified by $\rho(t)$. Lastly, $\eta_i(t) + \varepsilon_i(t)$ is the innovation term, which is further decomposed into two parts. The first part $\eta_i(t)$ captures the time dependence across $0 \leqslant t \leqslant 1$ while the second part $\varepsilon_i(t)$ is assumed to be i.i.d. As a result, without the network effect, the NFVC model will degenerate to a classic functional regression model (Yao

et al., 2005a; Chiou et al., 2003; Li and Hsing, 2010; Zhu et al., 2012). For the ith individual, we observe the nodal covariate $\boldsymbol{X}_i$ and corresponding responses at time points $\{t_{ik} : 1 \leqslant k \leqslant n_i\}$, i.e., $\{Y_i(t_{i1}), \ldots, Y_i(t_{in_i})\}$. We treat $\boldsymbol{X}_i$ as non-stochastic, hence we work in the fixed design context. The model can be easily generalized to incorporate functional covariates $\boldsymbol{X}_i(t)$, please see details in Section C.1 in the supplementary material. We also assume we observe the network connections a_{ij} for $i, j = 1, \ldots, N$.

Without loss of generality, we assume $\boldsymbol{X}_i$ is standardized, so $N^{-1}\sum_{i=1}^N \boldsymbol{X}_i = \mathbf{0}$ and $N^{-1}\sum_{i=1}^N \boldsymbol{X}_i\boldsymbol{X}_i^\top = \mathbf{I}_q$. Next, following the convention of functional regression (Yao et al., 2005a), we express $\eta_i(t)$ as a linear combination of functional principal components. Specifically, $\eta_i(t) = \sum_{k=1}^\infty b_{ik}\phi_k(t)$, where b_{ik}s are centered uncorrelated random variables with $E(b_{ik}^2) = \lambda_k$, and $\phi_k(t)$'s are the corresponding orthogonal eigenfunctions. Define $C(s,t) = \mathrm{cov}\{\eta_i(s), \eta_i(t)\}$. Then $C(s,t) = \sum_{k=1}^\infty \lambda_k\phi_k(s)\phi_k(t)$.

Following the convention (Hall and Hosseini-Nasab, 2006; Li and Hsing, 2010), we assume the nonzero λ_k's are distinct. In addition, $\eta_i(\cdot)$ and $\eta_j(\cdot)$ are independent of each other for $i \neq j$. Lastly, we assume $\{\varepsilon_i(t)\}$ to be an independent Gaussian process with $\mathrm{var}\{\varepsilon_i(t)\} = \sigma_\varepsilon^2$.

2.2 Notations

For convenience, we define the following notations. Let $\boldsymbol{Y}(t) = \{Y_1(t), \ldots, Y_N(t)\}^\top \in \mathbb{R}^N$, and $\boldsymbol{X} = (\boldsymbol{X}_1, \ldots, \boldsymbol{X}_N)^\top \in \mathbb{R}^{N\times q}$. Let $\mathbf{W} = (w_{ij})_{i,j=1}^N$ be the row normalized adjacency matrix, where $w_{ij} = m_i^{-1}a_{ij}$. In addition, define $\mathfrak{Y}(t) = \{\eta_1(t), \ldots \eta_N(t)\}^\top \in \mathbb{R}^N$, $\mathcal{E}(t) = \{\varepsilon_1(t), \cdots, \varepsilon_N(t)\}^\top \in \mathbb{R}^N$. Let $\varepsilon_i^*(t) = \eta_i(t) + \varepsilon_i(t)$, and $\mathcal{E}^*(t) = \{\varepsilon_1^*(t), \cdots, \varepsilon_N^*(t)\}^\top$. Then we have $\mathcal{E}^*(t) = \mathfrak{Y}(t) + \mathcal{E}(t)$, and $\sigma_\varepsilon^{*2}(t) \stackrel{\mathrm{def}}{=} \mathrm{var}\{\varepsilon_i^*(t)\} = C(t,t) + \sigma_\varepsilon^2$. Then (2.1)

can be expressed using a matrix form as

$$\boldsymbol{Y}(t) = \boldsymbol{X}\boldsymbol{\beta}(t) + \rho(t)\mathbf{W}\boldsymbol{Y}(t) + \mathcal{E}^*(t) = \boldsymbol{X}\boldsymbol{\beta}(t) + \rho(t)\mathbf{W}\boldsymbol{Y}(t) + \mathfrak{Y}(t) + \mathcal{E}(t). \quad (2.2)$$

Let $\mathbf{S}(t) = \mathbf{I}_N - \rho(t)\mathbf{W}$ and $\boldsymbol{Y}^*(t) = \mathbf{S}^{-1}(t)\boldsymbol{X}\boldsymbol{\beta}(t) + \mathbf{S}^{-1}(t)\mathfrak{Y}(t)$, then (2.2) further leads to $\mathbf{S}(t)\boldsymbol{Y}(t) = \boldsymbol{X}\boldsymbol{\beta}(t) + \mathfrak{Y}(t) + \mathcal{E}(t)$ and

$$\boldsymbol{Y}(t) = \boldsymbol{Y}^*(t) + \mathbf{S}^{-1}(t)\mathcal{E}(t). \quad (2.3)$$

We collect all parameter functions by $\boldsymbol{\theta}(t) = (\rho(t), \boldsymbol{\beta}(t)^\top)^\top \in \mathbb{R}^{q+1}$.

Remark 1. *The model (2.2) can be seen as a generalization of the autoregressive model in the time series analysis literature to the network/spatial data setting. To guarantee the model stationarity, we typically require* $\sup_t |\rho(t)| < 1$ *(Lee and Yu, 2009; Yu and Lee, 2010; Lee and Yu, 2013). As a result, we can express the stationary solution of (2.2) as* $\boldsymbol{Y}(t) = \mathbf{S}(t)^{-1}\boldsymbol{X}\boldsymbol{\beta}(t) + \sum_{k=0}^{\infty} \rho(t)^k \mathbf{W}^k \mathcal{E}^*(t)$.

For an arbitrary matrix $\mathbf{M} = (m_{ij}) \in \mathbb{R}^{p\times p}$, let $\lambda_i(\mathbf{M})$ be the ith eigenvalue of $\mathbf{M}$, where $|\lambda_1(\mathbf{M})| \geqslant |\lambda_2(\mathbf{M})| \geqslant \cdots \geqslant |\lambda_p(\mathbf{M})|$. In addition, let $\|\mathbf{M}\|_F = \mathrm{tr}(\mathbf{M}^\top\mathbf{M})^{1/2}$ be the Frobenius norm of the matrix. For any vector $\mathbf{v} \in \mathbb{R}^p$, let $\mathbf{v}_{-i} \in \mathbb{R}^{p-1}$ be the vector excluding the ith element. Let $\mathbf{e}_i$ be the vector with its ith element 1 and all others 0. For an arbitrary function $\phi(t)$ ($t \in [0,1]$), define $\|\phi\| = \{\int_0^1 \phi^2(t)dt\}^{1/2}$ and $\|\phi\|_\infty = \sup_{t\in[0,1]} |\phi(t)|$. For a vector function $\boldsymbol{\theta}(t) = (\theta_j(t))^\top$, let $\|\boldsymbol{\theta}\|_\infty = \max_j \|\theta_j(t)\|_\infty$. Lastly, denote $K(\cdot)$ as a standard kernel function and let $K_h(x) = h^{-1}K(x/h)$ with bandwidth h.

3 Functional Coefficients Estimation and Inference

3.1 A Least Squares Estimation Method

We first consider a hypothetical situation where $Y_i(t)$ is available at the same time t for all i. We can then rewrite (2.2) as $\boldsymbol{Y}(t) = \mathbf{S}(t)^{-1}\{\boldsymbol{X}\boldsymbol{\beta}(t) + \mathcal{E}^*(t)\}$, where $\mathbf{S}(t) = \mathbf{I}_N - \rho(t)\mathbf{W}$, $E\{\mathcal{E}^*(t)\} = \mathbf{0}$ and $\mathrm{var}\{\mathcal{E}^*(t)\} = \{C(t,t) + \sigma_\varepsilon^2\}\mathbf{I}_N = \sigma_\varepsilon^{*2}\mathbf{I}_N$. We point out two methods for estimating $\boldsymbol{\theta}(t)$ in this setting. The first is a quasi maximum likelihood estimator (Lee, 2004a), obtained by maximizing

$$\log|\mathbf{S}(t)| - \frac{N}{2}\log(\sigma_\varepsilon^{*2}) - \frac{1}{2\sigma_\varepsilon^{*2}}\big\{\mathbf{S}(t)\boldsymbol{Y}(t) - \boldsymbol{X}\boldsymbol{\beta}(t)\big\}^\top\big\{\mathbf{S}(t)\boldsymbol{Y}(t) - \boldsymbol{X}\boldsymbol{\beta}(t)\big\} \tag{3.1}$$

with respect to $\boldsymbol{\theta}(t)$. Although this estimator is statistically efficient and stable in many scenarios, it requires computing the determinant of $\mathbf{S}(t)$, which is an $N \times N$ matrix and hence the computation is expensive. A computationally more feasible alternative, proposed in Zhu et al. (2020) and Huang et al. (2019), is to minimize

$$Q\{\boldsymbol{\theta}(t)\} = \frac{1}{N}\Big\|\boldsymbol{D}(t)\mathbf{S}^\top(t)\big\{\mathbf{S}(t)\boldsymbol{Y}(t) - \boldsymbol{X}\boldsymbol{\beta}(t)\big\}\Big\|^2 \tag{3.2}$$

with respect to $\boldsymbol{\theta}(t)$, where $\boldsymbol{D}(t) = [\mathrm{diag}\{\mathbf{M}(t)\}]^{-1}$ and $\mathbf{M}(t) = \mathbf{I}_N + \rho^2(t)\mathbf{W}^\top\mathbf{W}$. To understand (3.2), we can verify that under normality of $\mathcal{E}^*(t)$, we could write $Q\{\boldsymbol{\theta}(t)\} = N^{-1}\sum_{i=1}^N \left[Y_i(t) - E\{Y_i(t)|\boldsymbol{Y}_{-i}(t)\}\right]^2$, where $\boldsymbol{Y}_{-i}(t)$ is the vector $\boldsymbol{Y}(t)$ with its ith component excluded. Hence we can view (3.2) as a composite maximum likelihood estimator of $\boldsymbol{\theta}(t)$, although its validity does not rely on the normal assumption.

Remark 2. *As pointed out in Zhu et al. (2020) and Huang et al. (2019), the main advantage of (3.2) is to consider both the first and second order friends in the compu-*

tation. To see this, we expand (3.2) as

$$Q\{\boldsymbol{\theta}(t)\} = \frac{1}{N}\sum_{i=1}^{N} d_i\{\rho(t)\}^2\Big\{Y_i(t) - \rho(t)\sum_j (w_{ji} + w_{ij})Y_j(t) + \sum_{j',j}\rho^2(t)w_{j'i}w_{j'j}Y_j(t) - \boldsymbol{X}_i^\top\boldsymbol{\beta}(t) + \sum_j \rho(t)w_{ji}\boldsymbol{X}_j^\top\boldsymbol{\beta}(t)\Big\}^2,$$

where $d_i\{\rho(t)\} = \{1 + \rho^2(t)\sum_{j=1}^N w_{ji}^2\}^{-1}$. *We observe that for each node* i, $Q\{\boldsymbol{\theta}(t)\}$ *involves only two types of nodes, the first order friends* $\{j : w_{ij} \neq 0 \text{ or } w_{ji} \neq 0\}$ *and the second order friends* $\{j : \sum_{j'=1}^N w_{j'i}w_{j'j} \neq 0\}$. *As a consequence, the computation is much less demanding especially when the network is sparse. Here, "sparse" means that the network density (i.e.,* $\sum_{i,j} a_{ij}/(N^2 - N)$*) is small, which implies fewer first- and second-order friends in general. Due to the computational simplicity, (3.2) will form the backbone of our proposed estimator.*

Remark 3. *In practice, the adjacency matrix* $\mathbf{A}$ *may vary with time* t *since the network connections may change over time, i.e.,* $\mathbf{A} = \mathbf{A}(t)$ *and* $\mathbf{W} = \mathbf{W}(t)$. *To handle this, we need to replace* $\mathbf{W}$ *with* $\mathbf{W}(t)$ *in the objective function (3.2). The subsequent estimation will then be slightly more complex since we need to smooth* $\mathbf{W}(t)$ *together with the responses. We provide the details of the estimation in Section C.2 of the supplementary material.*

3.2 Kernel-based Least Squares Estimator (KLSE)

Minimizing (3.2) is of course not feasible because at any t, we do not necessarily observe the whole vector $\boldsymbol{Y}(t)$. To overcome the hurdle that the observations are made at irregular time points for different individuals, we use the kernel idea to localize the individual terms in (3.2). Specifically, let $\mathbf{U}_i\{\rho(t)\} \equiv \boldsymbol{D}(t)\mathbf{S}^\top(t)\mathbf{S}(t)\mathbf{e}_i$ and $\boldsymbol{\xi}\{\boldsymbol{\theta}(t)\} \equiv$

$\boldsymbol{D}(t)\mathbf{S}^\top(t)\boldsymbol{X}\boldsymbol{\beta}(t)$. Then the objective function $Q\{\boldsymbol{\theta}(t)\}$ can be rewritten as

$$Q\big\{\boldsymbol{\theta}(t)\big\} = \frac{1}{N}\Bigg[\sum_{i,j=1}^{N}\mathbf{U}_i^\top\{\rho(t)\}\mathbf{U}_j\{\rho(t)\}Y_i(t)Y_j(t) \\ -2\sum_{i=1}^{N}\mathbf{U}_i\{\boldsymbol{\theta}(t)\}^\top\boldsymbol{\xi}\{\boldsymbol{\theta}(t)\}Y_i(t) + \boldsymbol{\xi}\{\boldsymbol{\theta}(t)\}^\top\boldsymbol{\xi}\{\boldsymbol{\theta}(t)\}\Bigg]. \tag{3.3}$$

Because $Y_i(t)$ may not be available, a natural idea is to replace $Y_i(t)$ by its neighbors using kernel method. We now investigate the details of this approach.

We first note that by (2.3) and the definition of $\boldsymbol{Y}^*(t)$, $\boldsymbol{Y}(t)$ can be decomposed into two parts, $\boldsymbol{Y}^*(t)$ and $\mathbf{S}^{-1}(t)\mathcal{E}(t)$, where $\boldsymbol{Y}^*(t)$ has time dependence along $t \in [0,1]$, while $\mathbf{S}^{-1}(t)\mathcal{E}(t)$ contains the pure noise. Note that $\boldsymbol{Y}^*(t)$ satisfies $\boldsymbol{Y}^*(t) = \boldsymbol{X}\boldsymbol{\beta}(t) + \rho(t)\mathbf{W}\boldsymbol{Y}^*(t) + \mathfrak{Y}(t)$, hence is a denoised version of $\boldsymbol{Y}(t)$. Now if we replace $Y_i(t)$'s with $Y_i^*(t)$'s in (3.3), we get

$$Q^*\{\boldsymbol{\theta}(t)\} = \frac{1}{N}\Big[\sum_{i,j=1}^{N}\mathbf{U}_i^\top\{\rho(t)\}\mathbf{U}_j\{\rho(t)\}Y_i^*(t)Y_j^*(t) \\ - 2\sum_{i=1}^{N}\mathbf{U}_i\{\boldsymbol{\theta}(t)\}^\top\boldsymbol{\xi}\{\boldsymbol{\theta}(t)\}Y_i^*(t) + \boldsymbol{\xi}\{\boldsymbol{\theta}(t)\}^\top\boldsymbol{\xi}\{\boldsymbol{\theta}(t)\}\Big], \tag{3.4}$$

which is a better target function to minimize than $Q\{\boldsymbol{\theta}(t)\}$ because $Q^*\{\boldsymbol{\theta}(t)\}$ is a denoised version of $Q\{\boldsymbol{\theta}(t)\}$. We now approximate $Y_i^*(t)Y_j^*(t)$ and $Y_i^*(t)$ by kernel smoothing. Specifically, let $Y_i^*(t)Y_j^*(t) \approx \nu_{h,ij}(t)/\widehat{f}_h^2(t)$ and $Y_i^*(t) \approx \zeta_{h,i}(t)/\widehat{f}_h(t)$, where

$$\nu_{h,ij}(t) = (n_i n_j)^{-1}\sum_{i'=1}^{n_i}\sum_{j'=1}^{n_j}Y_i(t_{ii'})Y_j(t_{jj'})K_h(t_{ii'}-t)K_h(t_{jj'}-t)I(i\neq j), \\ + \frac{1}{n_i^2-n_i}\sum_{i'\neq j'}Y_i(t_{ii'})Y_i(t_{ij'})K_h(t_{ii'}-t)K_h(t_{ij'}-t), \tag{3.5}$$

$$\zeta_{h,i}(t) = n_i^{-1}\sum_{i'=1}^{n_i}Y_i(t_{ii'})K_h(t_{ii'}-t) \tag{3.6}$$

and $\widehat{f}_h(t) = n^{-1}\sum_{i=1}^{N}\sum_{i'=1}^{n_i} K_h(t_{ii'} - t)$, where $n = \sum_{i=1}^{N} n_i$. Inserting these approximations into (3.4), we obtain the kernel smoothed objective function

$$\widetilde{Q}\{\boldsymbol{\theta}(t)\} = \widetilde{Q}_1\{\boldsymbol{\theta}(t)\} + \widetilde{Q}_2\{\boldsymbol{\theta}(t)\} + \widetilde{Q}_3\{\boldsymbol{\theta}(t)\}, \tag{3.7}$$

where

$$\begin{aligned}
\widetilde{Q}_1\{\boldsymbol{\theta}(t)\} &= \frac{1}{N}\sum_{i,j}^{N} \mathbf{U}_i\{\rho(t)\}^\top \mathbf{U}_j\{\rho(t)\}\frac{\nu_{h,ij}(t)}{\widehat{f}_h^2(t)},\\
\widetilde{Q}_2\{\boldsymbol{\theta}(t)\} &= \frac{-2}{N}\sum_{i=1}^{N} \mathbf{U}_i\{\rho(t)\}^\top \boldsymbol{\xi}\{\boldsymbol{\theta}(t)\}\frac{\zeta_{h,i}(t)}{\widehat{f}_h(t)},\\
\widetilde{Q}_3\{\boldsymbol{\theta}(t)\} &= \frac{1}{N}\boldsymbol{\xi}\{\boldsymbol{\theta}(t)\}^\top \boldsymbol{\xi}\{\boldsymbol{\theta}(t)\}.
\end{aligned}$$

We define the minimizer of (3.7), i.e. $\widehat{\boldsymbol{\theta}}(t) = \arg\min_{\boldsymbol{\theta}(t)} \widetilde{Q}\{\boldsymbol{\theta}(t)\}$, as the kernel-based least squares estimator (KLSE).

Remark 4. *Note that we especially exclude the square terms $Y_i(t_{ii'})^2$ in (3.5) because such terms will generate the error variance σ_ε^2, which should be zero in the denoised version $Q^*\{\boldsymbol{\theta}(t)\}$.*

3.3 Asymptotic Theory

We now show that the KLSE is consistent. To this end, we require a set of regularity conditions. The first two conditions are imposed on observations and functional coefficients. The third condition concerns the network structure. The last three conditions are related to the covariates and kernel estimation.

(C1) (Observation Distribution) The observation times $t_{ij}, i = 1, \ldots, N, j = 1, \ldots, n_i$, are i.i.d. with probability density function (pdf) $f(t)$. Further, $E\{|b_{ij}|^{2\lambda}\} <$

∞, $E\{b_{ij}^2 b_{ij'}^2\} = \lambda_j \lambda_{j'}$, $E\{b_{ij}^3 b_{ij'}\} = 0$ for $j \neq j'$, and $E\{|\varepsilon_i(t)|^{2\lambda}\} < \infty$ for some $\lambda > 4$.

(C2) (Function Bounds and Smoothness) The functions $f(t)$, $\rho(t)$, $\boldsymbol{\beta}(t)$, $\phi_k(t)$ are twice differentiable. In addition, the supreme norm of these functions, of their first and second order derivatives are all bounded by a finite constant. Furthermore $\inf_{t\in[0,1]} \min f(t) > 0$.

(C3) (Network Structure) The matrix $\mathbf{W}$ is non-stochastic.

(C3.1) (Connectivity) Treat $\mathbf{W}$ as a transition probability matrix of a Markov chain, whose state space is defined as the set of all the nodes in the network (i.e., $\{1, \cdots, N\}$). Assume the Markov chain is irreducible and aperiodic. Define $\pi = (\pi_1, \cdots, \pi_N)^\top \in \mathbb{R}^N$ as the stationary distribution of the Markov chain, such that (a) $\pi_i \geqslant 0$ and $\sum_{i=1}^N \pi_i = 1$, (b) $\pi = \mathbf{W}^\top \pi$, then $\sum_{i=1}^N \pi_i^2 = O(N^{-1/2-\delta})$, where $0 < \delta \leqslant 1/2$.

(C3.2) (Regularization) Let $\mathbb{W} = \mathbf{W} + \mathbf{W}^\top$, then $\max_i |\lambda_i(\mathbb{W})| = O(\log N)$ and $\max_i \mathbf{e}_i^\top \mathbb{W}^k \mathbf{1} = O\{(\log N)^{K_w}\}$ for integer $k = 1, \ldots, 4$, where $\lambda_i(\mathbb{W})$ is the ith eigenvalue of $\mathbb{W}$, and K_w is a finite positive integer.

(C3.3) (Convergence) The limits given in (A.4)–(A.7) of Section A.2 in the supplementary material exist.

(C4) (Covariates) Let $\mathbf{M}(t)$ be an arbitrary $N \times N$ matrix that satisfies $\sup_t \|\mathbf{M}(t)\|_F = o(k_N)$ when $N \to \infty$, where $k_N \to \infty$ when $N \to \infty$. Then

$$\frac{1}{k_N} \sup_t \left|(\boldsymbol{X}\boldsymbol{\beta}(t))^\top \mathbf{M}(t)(\boldsymbol{X}\boldsymbol{\beta}(t))\right| \leqslant \frac{c_x}{k_N} \sup_t \left|\mathrm{tr}\{\mathbf{M}(t)\}\right|$$

as $N \to \infty$, where $\|\boldsymbol{\beta}\|_\infty < \infty$ and c_x is a finite positive constant not related to

$\mathbf{M}$, N and k_N.

(C5) (Time Points and Bandwidth) Define $n_{\min} = \min_i n_i$. The bandwidth h satisfies $h^2 = o(N^{-1/2-\delta_1})$ and $n_{\min} h \geqslant N^{\delta_2}$, where δ_1 and δ_2 are finite positive constants satisfying $\delta_1 + \delta_2 < 1/2 - 2/\lambda$.

(C6) (Kernel Function) The kernel function $K(\cdot)$ is symmetric and of bounded variation, bounded and is supported on $[-1, 1]$.

(C7) (Identifiability Condition) Define $\boldsymbol{X}^*(t) = \{\mathbf{W}\mathbf{S}^{-1}(t)\boldsymbol{X}\boldsymbol{\beta}(t), \boldsymbol{X}\}$. There exist positive constants δ and τ such that

$$\min_{|\rho(t)|<1-\delta} \frac{1}{N} \boldsymbol{X}^{*\top}(t)\mathbf{S}(t)\mathbf{S}^{\top}(t)\boldsymbol{X}^*(t) \geqslant \tau > 0, \tag{3.8}$$

and $\rho(t) \in [-(1-\delta), 1-\delta]$ for all $t \in [0, 1]$.

Condition (C1) ensures that the observed time points are sufficiently and homogeneously spread out on the entire time domain instead of clumping in some regions. Condition (C2) requires related functions to be bounded and sufficiently smoothness. Condition (C3) puts various constraints on the network structure $\mathbf{W}$, all are routinely required in standard network literature (Zhu et al., 2017, 2020; Huang et al., 2020). Specifically, Condition (C3.1) requires the network to satisfy certain degree of connectivity. This is a weak condition and can easily hold for networks in reality (Newman et al., 2011). Condition (C3.2) restricts the extent of the heterogeneity of the nodes in the network. Particularly, $\mathbf{e}_i^\top \mathbb{W}^k \mathbf{1}$ can be treated as kth order degree of the node i. Hence, it restricts the "superstar" effects in the network. In addition, the weak network dependence is required by restricting the largest absolute eigenvalue of $\mathbb{W}$. Lastly, Condition (C3.3) requires certain convergence of network structures, which is

also assumed in the cross-sectional case (Zhu et al., 2020). Condition (C3.3) is also verified numerically in Section C of the supplementary materials.

Condition (C4) can be seen as a law of large number type assumption on $\boldsymbol{X}$ (Lee, 2004b; Zhu et al., 2020). Condition (C5) imposes requirements on the number of time points and the bandwidth. As outlined in Zhang and Wang (2016), we consider here an ultra-dense functional data scheme. This assures the parametric functional data convergence rate and leads to ignorable asymptotic bias. Condition (C5) and Condition (C3.2) lead to weak time and network dependences. In general, a larger N results in a smaller h, which then leads to larger number of time points (i.e., larger $n_{\min}$) to make a valid estimation (Zhu et al., 2012). Condition (C6) is a standard requirement on the kernel function. Lastly, Condition (C7) contains the identifiability requirement of the parameters, which restricts the multicollinearity effect of $\boldsymbol{X}^*(t)$. In addition, the network effect is restricted within $(-1, 1)$ to guarantee the stationarity of the process over the whole network (Liu, 2014; Yang and Lee, 2017; Zhu et al., 2020). Similar identifiability conditions are widely used in the literature (Zhu et al., 2020; Yang and Lee, 2017).

With the above preparation, we are able to establish the consistency and asymptotic normality of the estimator $\widehat{\boldsymbol{\theta}}(t)$.

Theorem 1. *Under Conditions (C1)–(C7), we have*

$$\sqrt{N}\Big(\widehat{\boldsymbol{\theta}}(t) - \boldsymbol{\theta}(t)\Big) \to_d N(0, \Sigma_{\boldsymbol{\theta}}(t)), \tag{3.9}$$

where $\Sigma_{\boldsymbol{\theta}} = \Sigma_2^{-1}(t)\Sigma_1(t)\Sigma_2^{-1}(t)$*, and* $\Sigma_1(t)$ *and* $\Sigma_2(t)$ *are defined in Section A.2 in the supplementary material.*

The proof is given in Section A.4 in the supplementary material. By Theorem 1, the

convergence rate $N^{-1/2}$ is slower than the standard nonparametric rate $(Nn_{\min}h)^{1/2}$, which is caused by the within-curve time dependence. Under weak dependence both in time and across network, the result of Theorem 1 shows that the estimator attains the parametric rate, and there is also no asymptotic bias, which agrees with the result under ultra-dense functional data discussed by Zhang and Wang (2016). In addition, the convergence rate here agrees with the theoretical result in the classical functional data case without network structure (Li and Hsing, 2010; Zhu et al., 2012).

4 Functional Principal Component Analysis (FPCA)

After conducting the parameter estimation, we proceed to perform the FPCA. The core part of FPCA is estimating the covariance function $C(s,t)$. In our notation, the spectral decomposition of $C(s,t)$ is

$$C(s,t) = \mathrm{cov}\{\eta_i(s), \eta_i(t)\} = \sum_{k=1}^{\infty} \lambda_k \phi_k(s)\phi_k(t), \tag{4.1}$$

where $\lambda_1 \geqslant \lambda_2 \geqslant \cdots \geqslant 0$ are ordered eigenvalues satisfying $\sum_{k=1}^{\infty} \lambda_k < \infty$, and $\phi_k(t)$'s are the corresponding orthonormal eigenfunctions. By the orthonormal property of the eigenfunctions, we have $\int C(s,t)\phi_k(s)ds = \lambda_k \phi_k(t)$. Using this relationship, once the estimation of the covariance function $C(s,t)$ is obtained, the eigenvalues and the eigenfunctions can be subsequently estimated by discretizing the smoothed covariance function; see discussions by Rice and Silverman (1991) and Capra and Müller (1997).

4.1 Estimation of the Covariance Function

To estimate the covariance function $C(s,t)$, a typical approach is to first estimate the individual function $\eta_i(t)$, denoted by $\widehat{\eta}_i(t)$, and then estimate $C(s,t)$ using $\widehat{C}(s,t) = (N-1)^{-1}\sum_{i=1}^{N}\widehat{\eta}_i(s)\widehat{\eta}_i(t)$ (Yao et al., 2005a; Zhu et al., 2012). However, this is infeasible in our setting when there is network dependence and different nodes are observed at different points. Specifically, to estimate $\eta_i(t)$, we typically use

$$\widehat{\eta}_i(t) = n_i^{-1}\sum_{k=1}^{n_i} K_h(t-t_{ik})\Big\{Y_i(t_{ik}) - \widehat{\rho}(t_{ik})m_i^{-1}\sum_{j=1}^{N} a_{ij}Y_j(t_{ik}) - \boldsymbol{X}_i^\top\widehat{\boldsymbol{\beta}}(t_{ik})\Big\}.$$

However, $Y_j(t_{ik})$'s are not generally available because we have allowed irregular and node-specific observation times.

We thus estimate $C(s,t)$ through an alternative approach. Note that $\mathrm{cov}\{\mathcal{E}^*(s),\mathcal{E}^*(t)\} = \mathrm{cov}\{\mathbf{S}(s)\boldsymbol{Y}(s) - \boldsymbol{X}\boldsymbol{\beta}(s), \mathbf{S}(t)\boldsymbol{Y}(t) - \boldsymbol{X}\boldsymbol{\beta}(t)\} = \{C(s,t) + \sigma_\varepsilon^2 I(s=t)\}\mathbf{I}_N$. Therefore

$$\begin{aligned}
& C(s,t) + \sigma_\varepsilon^2 I(s=t) \\
= \; & \frac{1}{N}\sum_i \mathrm{cov}\Big\{\sum_{j=1}^{N}\mathbf{S}_{ij}(s)Y_j(s) - \boldsymbol{X}_i^\top\boldsymbol{\beta}(s), \sum_{k=1}^{N}\mathbf{S}_{ik}(t)Y_k(t) - \boldsymbol{X}_i^\top\boldsymbol{\beta}(t)\Big\} \\
= \; & \frac{1}{N}\sum_{i=1}^{N}\sum_{j,k=1}^{N}\mathbf{S}_{ij}(s)\mathbf{S}_{ik}(t)E\{Y_j(s)Y_k(t)\} - \frac{1}{N}\boldsymbol{\beta}^\top(s)\boldsymbol{X}^\top\boldsymbol{X}\boldsymbol{\beta}(t).
\end{aligned}$$

Because our goal is in estimating $C(s,t)$ without the additional error variance term $\sigma_\varepsilon^2 I(s=t)$, we take advantage of the uncorrelated error property, and directly exclude the possibility of generating error variance σ^2 in approximating $E\{Y_j(s)Y_k(t)\}$. Specifically, let

$$\hat{v}_{jk}(s,t) = \frac{\sum_{j'=1}^{n_j}\sum_{k'=1}^{n_k} K_h(t_{jj'}-s)K_h(t_{kk'}-t)Y_j(t_{jj'})Y_k(t_{kk'})I\{(j,j')\neq(k,k')\}}{\sum_{j'=1}^{n_j}\sum_{k'=1}^{n_k} K_h(t_{jj'}-s)K_h(t_{kk'}-t)I\{(j,j')\neq(k,k')\}}.$$

Because the summation in $\hat{v}_{jk}(s,t)$ does not include the case $(j,j') = (k,k')$, it eliminates σ_ε^2 explicitly. Then we estimate $C(s,t)$ by

$$\widehat{C}(s,t) = \frac{1}{N}\sum_{i=1}^{N}\sum_{j,k=1}^{N} \widehat{\mathbf{S}}_{ij}(s)\widehat{\mathbf{S}}_{ik}(t)\hat{v}_{jk}(s,t) - \frac{1}{N}\widehat{\boldsymbol{\beta}}^\top(s)\boldsymbol{X}^\top\boldsymbol{X}\widehat{\boldsymbol{\beta}}(t), \tag{4.2}$$

where $\widehat{\mathbf{S}}_{ij}(t) = \{\mathbf{I}_N - \widehat{\rho}(t)\mathbf{W}\}_{(i,j)}$.

To establish the convergence result of $\widehat{C}(s,t)$, we first establish a uniform convergence result for $\widehat{\boldsymbol{\theta}}(t)$, which strengthens the corresponding result in Theorem 1.

Theorem 2. *Under Conditions (C1)–(C7),*

$$\sup_t \|\widehat{\boldsymbol{\theta}}(t) - \boldsymbol{\theta}(t)\| = O_p\{h^2(\log N)^6 + \delta_N^{1/2}\}, \tag{4.3}$$

where $\delta_N = (\log N)^{\max(K_w,10)+3}/N$.

The proof of Theorem 2 is given in Section A.5 in the supplementary material. Note that the convergence rate of $\sup_t \|\widehat{\boldsymbol{\theta}}(t) - \boldsymbol{\theta}(t)\|$ is slightly slower than that in Theorem 1, which agrees with the result in the independence case (Zhu et al., 2012). Moreover, the convergence rate is also related to the network dependence. Specifically, a higher extent of heterogeneity (i.e., larger K_w) will result in a slower convergence rate. Using Theorem 2, we are able to establish the uniform convergence for $\widehat{C}(s,t)$.

Theorem 3. *Under Conditions (C1)–(C7),*

$$\sup_{s,t} \left|\widehat{C}(s,t) - C(s,t)\right| = O_p\{h^2(\log N)^6 + \delta_N^{1/2}\}, \tag{4.4}$$

where $\delta_N = (\log N)^{\max(K_w,10)+3}/N$.

The proof of Theorem 3 is given in Section A.6 in the supplementary material.

Compared to the independent case (Li and Hsing, 2010; Zhu et al., 2012), the convergence rate of $\widehat{C}(s,t)$ is also slower due to the existence of the network dependence effect. Because of the network dependence, which propagates to all aspects of the proof, in combination with the time correlation within each node, Theorem 2 and Theorem 3 are very difficult to establish. The proofs involve empirical process treatments, nonparametric analysis, and various concentration inequalities on martingale sequences. In fact, a Bernstein's type inequality is further established under network condition (C3) in Lemma 6 and Lemma 10 in the supplementary material, which may be of independent interest.

4.2 Convergence of FPCA

Once the uniform convergence result is established for $\widehat{C}(s,t)$, we proceed to analyze the convergence of FPCA by using (4.1). Note that $\phi_j(t)$ and $\widehat{\phi}_j(t)$ are identifiable up to a sign change. Following the convention (Hall and Hosseini-Nasab, 2006), we allow $\phi_j(t)$ to take an arbitrary sign but we choose the sign of $\widehat{\phi}_j$ such that $\|\widehat{\phi}_j - \phi_j\|$ is minimized. We establish the convergence of the eigen-decomposition in Corollary 1.

Corollary 1. *Assume Conditions (C1)–(C7) to hold. Let J_0 be a positive constant and let $\lambda_1 > \lambda_2 > \cdots \lambda_j > \cdots$. In addition, for $1 \leqslant j < k \leqslant J_0$, assume $\min_{\lambda_j,\lambda_k>0}(\lambda_j - \lambda_k) \geqslant C > 0$. Let δ_N be the same as in Theorem 3. Then for any j $(1 \leqslant j \leqslant J_0)$,*
(a) $\widehat{\lambda}_j - \lambda_j = O_p\{h^2(\log N)^6 + \delta_N^{1/2}\}$;
(b) $\|\widehat{\phi}_j - \phi_j\| = O_p\{h^2(\log N)^6 + \delta_N^{1/2}\}$;
(c) $\sup_t \|\widehat{\phi}_j(t) - \phi_j(t)\| = O_p\{h^2(\log N)^6 + \delta_N^{1/2}\}$.

Compared to the conventional approaches, the FPC analysis is conducted under the effects of both exogenous covariates $\boldsymbol{X}$ and network dependence. Similar to the

conclusion in the independent case, the convergence rate is the same as the estimated covariance function in (4.4).

To summarize, the theoretical contributions in this paper are mainly in three aspects. First, we establish the uniform convergence of the functional model parameter $\boldsymbol{\theta}(t)$ under the network dependence structure. Second, we establish the uniform convergence of the covariance function by incorporating the network dependence measure. Third, we establish the convergence of eigenvalues and eigenfunctions, which further leads to convergence of the functional principal component scores. As a result, this enables us to perform predictions, which are discussed in detail below.

4.3 Estimation of FPC Scores and Selection of Number of Eigenfunctions

Following the standard literature (Yao et al., 2005a), assume that the process $\eta_i(t)$ can be well approximated by using the first K eigenfunctions, i.e., $\eta_i(t) \approx \sum_{k=1}^{K} b_{ik}\phi_k(t)$. The eigenfunctions $\phi_k(t)$ is estimated by the eigen-decomposition of $\widehat{C}(s,t)$ and the FPC scores b_{ik} is subsequently obtained.

Specifically, define the grid points equally spaced within $[0,1]$ as $v_1 < v_2 < \cdots < v_{\tilde{n}}$. Let $\boldsymbol{\eta}_i = (\eta_i(v_1), \cdots, \eta_i(v_{\tilde{n}}))^\top \in \mathbb{R}^n$ and $\boldsymbol{\phi}_k = (\phi_k(v_1), \cdots, \phi_k(v_{\tilde{n}}))^\top \in \mathbb{R}^{\tilde{n}}$. Following Yao et al. (2005a), the FPC scores b_{ik} can be estimated using conditional expectation on $\boldsymbol{\eta}_i$ as $\widetilde{b}_{ik} = E(b_{ik}|\boldsymbol{\eta}_i) = \lambda_k \boldsymbol{\phi}_k^\top \Sigma_{\boldsymbol{\eta}_i}^{-1} \boldsymbol{\eta}_i$, where $\Sigma_{\boldsymbol{\eta}_i} = \mathrm{cov}(\boldsymbol{\eta}_i, \boldsymbol{\eta}_i)$. The trajectory $\eta_i(t)$ that we aim to predict can be expressed as $\eta_i^*(t) = \sum_{k=1}^{\infty} \widetilde{b}_{ik}\phi_k(t)$. In practice, we estimate b_{ik} as as $\widehat{b}_{ik} = \widehat{\lambda}_k \widehat{\boldsymbol{\phi}}_k^\top \widehat{\Sigma}_{\boldsymbol{\eta}_i}^{-1} \mathcal{Y}_i$, where $\widehat{\Sigma}_{\boldsymbol{\eta}_i} = (\widehat{C}(v_k, v_l))_{1 \leqslant k \leqslant l \leqslant \tilde{n}} \in \mathbb{R}^{\tilde{n} \times \tilde{n}}$ is obtained

from (4.2), $\mathcal{Y}_i = (\mathcal{Y}_{i1}, \cdots, \mathcal{Y}_{in})^\top$, where

$$\mathcal{Y}_{ik} = \zeta_{h,i}(v_k)/\widehat{f}_h(v_k) - \widehat{\rho}(v_k) m_i^{-1} \sum_j a_{ij} \zeta_{h,j}(v_k)/\widehat{f}_h(v_k) - \boldsymbol{X}_i^\top \widehat{\boldsymbol{\beta}}(v_k),$$

and $\zeta_{h,i}(v_k)$ given in (3.6). As a result, we are able to predict $\eta_i(t)$ by using the first K eigenfunctions as $\widehat{\eta}_i^K(t) = \sum_{k=1}^K \widehat{b}_{ik}\widehat{\phi}_k(t)$. We establish the theoretical properties of $\widehat{b}_{ik}$ and $\widehat{\eta}_i^K(t)$ in Corollary 2.

Corollary 2. *Assume Conditions (C1)–(C7) and $\widetilde{n} = O(1)$. Then $\lim_{N\to\infty} \widehat{b}_{ik} = \widetilde{b}_{ik}$ with probability tending to 1. In addition, for all $t \in (0,1)$, $\lim_{K\to\infty}\lim_{N\to\infty} \widehat{\eta}_i^K(t) = \eta_i^*(t)$ in probability.*

The proof of Corollary 2 is provided in Section A.8 in the supplementary material . By Corollary 2, the predictor $\widehat{\eta}_i^K(t)$ behaves as well as the best linear estimator $\eta_i^*(t)$. To select the number of eigenfunctions K, we can adopt AIC and BIC type criteria (Yao et al., 2005a). The profiled pseudo-Gaussian log-likelihood given all estimated FPC scores $\widehat{b}_{ik}$ is

$$\widehat{\ell}(K) = \widetilde{n}^{-1} \sum_{l=1}^{\widetilde{n}} \Big[\log|\widehat{\mathbf{S}}(v_l)| - \frac{N}{2} \log\Big\{ \Big\| \widehat{\mathbf{S}}(v_l)\widetilde{\zeta}_h(v_l) - \boldsymbol{X}\widehat{\boldsymbol{\beta}}(v_k) - \sum_{k=1}^K \widehat{\mathbf{b}}_k \widehat{\phi}_k(v_l) \Big\|^2 \Big\} \Big],$$

where $\widetilde{\zeta}_{h,i}(t) = \zeta_{h,i}(t)/\widehat{f}_h(t)$, $\widetilde{\zeta}_h(t) = (\widetilde{\zeta}_{h,1}(t), \cdots, \widetilde{\zeta}_{h,N}(t))^\top$, $\boldsymbol{w}_i = (w_{i1}, \cdots, w_{iN})^\top$, and $\widehat{\mathbf{b}}_k = (\widehat{b}_{1k}, \cdots, \widehat{b}_{Nk})^\top$. Then the AIC and BIC objective functions are $\ell_{\mathrm{AIC}}(K) = -2\widehat{\ell}(K) + 2K$ and $\ell_{\mathrm{BIC}}(K) = -2\widehat{\ell}(K) + KC_N$, where C_N is a function of N and $C_N/N \to 0$ (Zhu et al., 2010; Cai et al., 2020). In practice, we recommend choosing C_n as $N^{0.8}$, which works very well in our simulations. The number of eigenfunctions are then selected by minimizing the objective functions as $K_{\mathrm{AIC}} = \arg\min_K \ell_{\mathrm{AIC}}(K)$ and $K_{\mathrm{BIC}} = \arg\min_K \ell_{\mathrm{BIC}}(K)$. Importantly, during this process, the first additive term can be discarded so we do not need to compute the determinant of the large matrix $\widehat{\mathbf{S}}(v_l)$.

5 NUMERICAL STUDIES

5.1 Simulation Models

We conducted extensive numerical studies to examine the finite sample performance of the proposed method. Two different network structures are considered: the stochastic block network and the power-law distributed network. In terms of the functional coefficients, we set the network effect $\rho(t) = 0.2\sin(2\pi t) + 0.3$, where $t \in [0, 1]$, and let the regression coefficient functions be $\beta_1(t) = 0.5t + 0.5$, $\beta_2(t) = 4(t - 0.5)^2 + 0.5$. Following the functional data setting in Zhu et al. (2012), we set the two eigenfunctions to be $\phi_1(t) = \sqrt{2}\cos(2\pi t)$, $\phi_2(t) = \sqrt{2}\sin(2\pi t)$, with corresponding eigenvalues $\lambda_1 = 1.4$, $\lambda_2 = 0.8$, and $\lambda_j = 0$ for $j \geqslant 3$.

Under these parameter settings, we generate the responses as follows. First we partition $[0, 1]$ into equally spaced grids $\mathcal{V} = \{v_1, v_2, \ldots, v_{n_{\max}}\}$ with $v_1 = 0$ and $v_{n_{\max}} = 1$. On the grids, we generate the responses $Y_i(v_j)$ for $1 \leqslant i \leqslant N$ and $1 \leqslant j \leqslant n_{\max}$ using the NFVC model (2.2). Specifically, the covariates $\boldsymbol{X}_i$ and the measurement errors $\varepsilon_i(t)$ are independently sampled from $N(\mathbf{0}, \mathbf{I}_2)$ and $N(0, \sigma^2)$ respectively, where $\sigma = 0.2$. Next, for the ith node, we randomly delete n_i^* responses on the grids, where the number n_i^* is uniformly sampled from $0 \leqslant n_i^* \leqslant 10$. As a result, the number of observed responses for each node grows in the same order as $n_{\max}$. Lastly, we describe the generating mechanisms of the two types of networks below, and also describe an additional experiment to study the network sparsity and its impact.

Scenario 1: (Stochastic Block Model) We first consider the stochastic block model, which is one of the most popular network structures in the literature (Wang and Wong, 1987; Nowicki and Snijders, 2001). To generate the model, we randomly assign a block label k ($k = 1, 2, \ldots, K$) for each node, where K is set to be 10. Let

$\text{pr}(a_{ij} = 1) = 0.9N^{-1}$ if i and j belong to the same block, and $\text{pr}(a_{ij} = 1) = 0.3N^{-1}$ otherwise. Consequently, nodes with the same block labels are more likely to be connected.

Scenario 2: (Power-Law Distributed Network) We next consider the power-law distributed network, which models the situation where the majority of nodes have relatively few links but a small proportion (i.e., celebrities) possess a large number of links (Barabási and Albert, 1999). The model assumes that the degrees follow the power-law distribution (Clauset et al., 2009). Specifically, we generate the in-degree $m_i = \sum_j a_{ij}$ for node i from the discrete power-law distribution, i.e., $P(m_i = k) = ck^{-\alpha}$, where c is the normalizing constant and $\alpha = 2.5$. Then we randomly select m_i nodes for the i-th node as its followers.

Scenario 3: (Network Sparsity) In this scenario, we evaluate the model performance under different levels of network sparsity. Specifically, we consider a dyad independence model (Holland and Leinhardt, 1981). A dyad is defined as $D_{ij} = (a_{ij}, a_{ji})$. The dyad independence model specifies the connection probability of different types of dyads. Specifically, we set $P\{D_{ij} = (1,1)\} = \alpha N^{-1}$ and $P\{D_{ij} = (1,0)\} = P\{D_{ij} = (0,1)\} = \alpha N^{-\nu}$ with $0 < \nu \leqslant 1$. Thus, the expected degree of each node is $O(\alpha N^{1-\nu})$, and $P\{D_{ij} = (0,0)\} = 1 - \alpha N^{-1} - 2\alpha N^{-\nu}$, which is close to 1 for large N. We vary (α, ν) from $(2, 0.9)$ to $(10, 0.5)$ to reflect different levels of network sparsity.

5.2 Performance Measurements and Simulation Results

To gauge a reliable evaluation of the simulation study, we repeat the experiments for 500 replications. The network size is set to be $N = 100, 200, 500$. with $n_{\max} = 50, 100, 200$ respectively. For simplicity, we select the bandwidth by the rule of thumb (Li and

Racine, 2007). To estimate the random functions, we follow the standard functional data estimation procedure in Yao et al. (2005a).

We first evaluate the estimation accuracy of the functional coefficients for Scenario 1–2. Similar to Yao et al. (2005b), we consider the integrated mean squared error (IMSE) for the curves, which is defined as $\sum_{i=1}^{N}\int_0^1\{\widehat{\theta}_{i,j}(t)-\theta_{i,j}(t)\}^2dt/N$ when estimating $\theta_j(t)$. We similarly define the IMSEs for the covariance function and the eigen-functions, and the mean squared error (MSE) for the eigenvalues. We also use supremum norm (SUP) to evaluate the performance of the functional estimators, i.e., $\mathrm{SUP}_{\theta_j} = \max_{1\leqslant k\leqslant R}\{\widehat{\theta}_j(t_k)-\theta_j(t_k)\}^2$, where we discretize the interval $[0,1]$ into 20 equally spaced intervals to facilitate the analysis. The SUPs for $\widehat{C}(s,t)$ and $\widehat{\phi}_j(t)$ are defined similarly.

The simulation results are summarized in Table 3. As we can see, as the network size N increases, the estimation accuracy grows higher. For instance, as N increases from 100 to 500, the IMSE for $\widehat{\rho}(t)$ drops from 1.31×10^{-3} to 0.026×10^{-3} for the Power-law distributed network. The estimated parameter functions are given in Figures E.1 and E.2, and the estimated eigenfunctions are presented in Figure E.3, all of which are in the online supplementary material. Likewise, as the sample size increases, the estimated parameters are closer to the true values and the estimation tends to be more stable with narrower confidence bands. This corroborates with the theoretical findings. To see the computational performance in Scenario 3. We vary (α,ν) from $(2,0.9)$ to $(10,0.5)$. Accordingly, the network density increases from 0.018 to 0.384. As we can observe from Figure E.4, the computation time (measured in seconds) is increasing as the network becomes denser, where we set $N=200$ and $n_{\max}=100$.

We further compare the prediction accuracy of the proposed method with the network autoregression (NAR) model in Zhu et al. (2017). Since the NAR model is a

Table 1: The average prediction errors for the stochastic block network and the power-law distributed network for different sizes of $(N, n_{\max})$.

		SBM		Power-law	
N	n_{max}	NFVC	NAR	NFVC	NAR
100	50	0.219	0.466	0.192	0.393
200	100	0.087	0.487	0.074	0.418
500	200	0.046	0.501	0.041	0.432

time series model and not built for functional data, we first smooth the observations for each subject. Subsequently, we split the time interval $[0, 1]$ into 20 equal length sub intervals and choose the median point in each interval as the observation time. Then, we use the grids $(1, 3, 5, \ldots, 19)$ for training, and test the prediction power on grids $(2, 4, \ldots, 20)$ for both methods. We summarize the average prediction errors for the stochastic block network and the power-law distributed network in Table 1. The prediction error is calculated as the IMSE, and the BIC criterion is used for selecting the number of principal components. Compared to Zhu et al. (2017), the proposed NFVC model clearly has higher prediction accuracy. In addition, the prediction error of the NFVC model reduces as the number of observations increases.

Lastly, we investigate the performance of AIC and BIC in determining K in Table 2. As expected, AIC always chooses a larger number of eigenfunctions compared to BIC. To aim for a parsimonious model, we recommend using BIC. We also perform sensitivity analysis for different choice of K. Specifically, we evaluate the prediction errors of our method by increasing K from 1 to 6, as demonstrated in Table 2. It can be seen clearly that the prediction error is not sensitive to the choice of K for $K \geqslant 2$.

Table 2: The average of K chosen by the AIC and BIC criterion and the average prediction errors for the stochastic block network and the power-law distributed network under the L_2 loss. K increases from 1 to 6. $N = 100$ and $n_{max} = 50$.

K	AIC	BIC	1	2	3	4	5	6
SBM	4.78	2.41	0.2303	0.2191	0.2154	0.2138	0.2130	0.2127
Power-law	4.88	2.35	0.2148	0.1927	0.1923	0.1920	0.1920	0.1921

6 Social Network Data Analysis

We now use the tools developed so far to analyze the behavior of users on social media using a social network dataset. The data is collected from Sina Weibo, which is similar to Twitter and is the largest online social media in mainland China. Our data set contains the posting behavior of $N = 850$ active users from Jan. 2nd, 2014 to Jan. 9th, 2014. Hence the time period includes a week starting on a Thursday.

The response $Y_i(t_{ik})$ is the number of posts of the ith user in the kth hour. $Y_i(t)$ reflects the activeness of the ith social network user, which is assumed to have a smooth trajectory of time t. As an example, Figure E.5 in the supplementary file contains a snapshot of two Weibo posts from one user in the data set. We also select one user "BreakingNews" and two of his following friends. All three users have backgrounds in social media and economics. The posting times of the three users are visualized in Figure 1, who tend to post more frequently during daytime. In addition, the active periods of the three users are similar, which indicates the user may share similar dynamic patterns with his connected friends. Next, the total number of posts by each user is illustrated in the histogram in the left panel of Figure 2, where the average post number per user is 146.7. We also draw the total number of Weibo posts by all the 850 users within each hour in the right panel of Figure 2. We can observe an obvious periodic pattern with roughly three peaks in each day. Besides the responses, we also

include two nodal covariates of the users. The first is gender, which is 1 for a male user and 0 otherwise. About 76% of the users are male. The second is the number of user self-created labels, which are short keywords defined by the user themselves to describe his/her life status. We construct the adjacency matrix by using users' follower-followee relationships, i.e., $a_{ij} = 1$ if the ith user follows the jth user on the Weibo platform. The network density is calculated as $\sum_{i,j} a_{ij}/\{N(N-1)\} = 0.047$. This implies that the network is highly sparse.

We used the NFVC model to analyze the data and the estimated functional coefficients are given in Figure 3. First, we can observe a daily periodic pattern for all estimates, where the estimated functions reach peaks in the day and drop down at night. The pattern reflects human being's diurnal nature. Second, the estimated network effect $\widehat{\rho}(t)$ is positive, which implies that a user's posting behavior is positively related to their following friends. The network strength seems to be much lower on Saturday, possibly because many people devote Saturday to aspects of life other than social network. Regarding the nodal covariates, $\widehat{\beta}_1(t)$ suggests that the male users tend to be less active than the female users on weekends, but more active on Thursday and Friday. The posting behavior of the two genders are more similar on other days. Finally, from $\widehat{\beta}_2(t)$, we find that the users with more self-created labels tend to be more active during the week days but less active on weekend. This is possibly the result of their more active personal characters, in combination with richer and more diversified personal lives on weekends.

We further conduct the FPCA analysis using the estimated $\widehat{C}(s,t)$. The BIC criterion sets $K = 3$. The estimated eigenfunctions are visualized in Figure 4. The first eigenfunction appears to reflect the general periodic dynamic patterns of user activities, while the second and third eigenfunctions seem to characterize the movements

during weekdays and weekends respectively.

We compare the prediction power of the proposed method with the method in Zhu et al. (2017). Specifically, we divide the time interval into 40 equal length sub-intervals, and use the grids $(1, 3, \ldots, 39)$ for model training and $(2, 4, \ldots, 40)$ for model testing. The number of eigenfunctions is selected by the BIC criterion, which leads to $K = 3$. The mean prediction errors measured in IMSE for NAR and NFVC models are 1.303 and 0.448 respectively. This indicates that the proposed method can capture the dynamic patterns in the network more accurately than Zhu et al. (2017), due to its ability to take into the functional nature of the user activeness.

7 Conclusion

In this article, we propose the NFVC model for analyzing functional data with known network structure information. Many extensions are possible following our work, which can potentially generate interesting research topics for future studies. For example, we only consider a dense sampling scheme for time points in this work in the theoretical discussions. A natural extension is to consider a sparse data setting (Yao et al., 2005a). One may also extend our univariate response case to a multivariate response analysis (Zhu et al., 2012). To this end, one may need to model the network effects and regression relationships among several indices of interest. Next, in this work we treat $\mathbf{W}$ as static. As an interesting extension, one may consider dynamic network relationships as discussed in Section C.2 in the supplementary material . Furthermore, time varying exogenous variables can be employed to enrich the dynamic network structures. Lastly, in this work, we only consider one type of network relationship in the model. In practice, the network users can be connected through more than one type of relationships.

Consequently, it will be of interest to consider evaluating multiple network effects in the model.

References

Acemoglu, D., Ozdaglar, A., and Tahbaz-Salehi, A. (2015), "Systemic risk and stability in financial networks," *American Economic Review*, 105, 564–608.

Barabási, A. and Albert, R. (1999), "Emergence of scaling in random networks," *Science*, 286, 509–512.

Cai, Z., Li, R., and Zhu, L. (2020), "Online Sufficient Dimension Reduction Through Sliced Inverse Regression." *Journal of Machine Learning Research*, 21, 1–25.

Capra, W. B. and Müller, H.-G. (1997), "An accelerated-time model for response curves," *Journal of the American Statistical Association*, 92, 72–83.

Chen, X., Chen, Y., and Xiao, P. (2013), "The impact of sampling and network topology on the estimation of social intercorrelation," *Journal of Marketing Research*, 50, 95–110.

Chiou, J.-M. (2012), "Dynamical functional prediction and classification, with application to traffic flow prediction," *The Annals of Applied Statistics*, 6, 1588–1614.

Chiou, J.-M., Müller, H.-G., and Wang, J.-L. (2003), "Functional quasi-likelihood regression models with smooth random effects," *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 65, 405–423.

Clauset, A., Shalizi, C., and Newman, M. (2009), "Power-law distributions in empirical data," *SIAM Review*, 51, 661–703.

Faraway, J. J. (1997), "Regression analysis for a functional response," *Technometrics*, 39, 254–261.

Ferraty, F., Van Keilegom, I., and Vieu, P. (2012), "Regression when both response and predictor are functions," *Journal of Multivariate Analysis*, 109, 10–28.

Hall, P. and Hosseini-Nasab, M. (2006), "On properties of functional principal components analysis," *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 68, 109–126.

Härdle, W. K., Wang, W., and Yu, L. (2016), "TENET: Tail-event driven network risk," *Journal of Econometrics*, 192, 499–513.

Holland, P. W. and Leinhardt, S. (1981), "An exponential family of probability distributions for directed graphs," *Journal of the american Statistical association*, 76, 33–50.

Huang, D., Lan, W., Zhang, H. H., Wang, H., et al. (2019), "Least squares estimation of spatial autoregressive models for large-scale social networks," *Electronic Journal of Statistics*, 13, 1135–1165.

Huang, D., Wang, F., Zhu, X., and Wang, H. (2020), "Two-mode network autoregressive model for large-scale networks," *Journal of Econometrics*.

Huang, M., Li, R., Wang, H., and Yao, W. (2014), "Estimating mixture of Gaussian processes by kernel smoothing," *Journal of Business & Economic Statistics*, 32, 259–270.

Lee, L. (2004a), "Asymptotic distributions of quasi-maximum likelihood estimators for spatial autoregressive models," *Econometrica*, 72, 1899–1925.

Lee, L.-F. (2004b), "Asymptotic distributions of quasi-maximum likelihood estimators for spatial autoregressive models," *Econometrica*, 72, 1899–1925.

Lee, L.-f. and Yu, J. (2009), "Spatial nonstationarity and spurious regression: The case with a row-normalized spatial weights matrix," *Spatial Economic Analysis*, 4, 301–327.

— (2013), "Near unit root in the spatial autoregressive model," *Spatial Economic Analysis*, 8, 314–351.

Leng, X. and Müller, H.-G. (2006), "Classification using functional data analysis for temporal gene expression data," *Bioinformatics*, 22, 68–76.

Li, Q. and Racine, J. S. (2007), *Nonparametric econometrics: theory and practice*, Princeton University Press.

Li, Y. and Hsing, T. (2010), "Uniform convergence rates for nonparametric regression and principal component analysis in functional/longitudinal data," *The Annals of Statistics*, 38, 3321–3351.

Liu, X. (2014), "Identification and efficient estimation of simultaneous equations network models," *Journal of Business & Economic Statistics*, 32, 516–536.

Müller, H.-g. (2005), "Functional modelling and classification of longitudinal data," *Scandinavian Journal of Statistics*, 32, 223–240.

Müller, H.-G. and Yao, F. (2008), "Funcional additive models," *Journal of the American Statistical Association*, 103, 1534–1544.

Newman, M., Barabasi, A.-L., and Watts, D. J. (2011), *The structure and dynamics of networks*, Princeton University Press.

Nowicki, K. and Snijders, T. A. B. (2001), "Estimation and prediction for stochastic blockstructures," *Journal of the American Statistical Association*, 96, 1077–1087.

Ramsay, J. and Dalzell, C. (1991), "Some tools for functional data analysis," *Journal of the Royal Statistical Society: Series B (Methodological)*, 53, 539–572.

Ramsay, J. O. and Silverman, B. W. (2007), *Applied functional data analysis: methods and case studies*, Springer.

Rice, J. A. and Silverman, B. W. (1991), "Estimating the mean and covariance structure nonparametrically when the data are curves," *Journal of the Royal Statistical Society: Series B (Methodological)*, 53, 233–243.

Sojourner, A. (2013), "Identification of peer effects with missing peer data: Evidence from Project STAR," *The Economic Journal*, 123, 574–605.

Wang, J.-L., Chiou, J.-M., and Müller, H.-G. (2016), "Functional data analysis," *Annual Review of Statistics and Its Application*, 3, 257–295.

Wang, Y. J. and Wong, G. Y. (1987), "Stochastic blockmodels for directed graphs," *Journal of the American Statistical Association*, 82, 8–19.

Yang, K. and Lee, L.-f. (2017), "Identification and QML estimation of multivariate and simultaneous equations spatial autoregressive models," *Journal of Econometrics*, 196, 196–214.

Yao, F., Müller, H.-G., and Wang, J.-L. (2005a), "Functional data analysis for sparse longitudinal data," *Journal of the American Statistical Association*, 100, 577–590.

— (2005b), "Functional data analysis for sparse longitudinal data," *J. Am. Statist. Assoc.*, 100, 577–590.

— (2005c), "Functional linear regression analysis for longitudinal data," *The Annals of Statistics*, 33, 2873–2903.

Yu, J. and Lee, L.-f. (2010), "Estimation of unit root spatial dynamic panel data models," *Econometric Theory*, 1332–1362.

Zhang, X. and Wang, J.-L. (2016), "From sparse to dense functional data and beyond," *The Annals of Statistics*, 44, 2281–2321.

Zhao, Y., Levina, E., and Zhu, J. (2012), "Consistency of community detection in networks under degree-corrected stochastic block models," *Annals of Statistics*, 40, 2266–2292.

Zhu, H., Li, R., and Kong, L. (2012), "Multivariate varying coefficient model for functional responses," *The Annals of statistics*, 40, 2634.

Zhu, L.-P., Zhu, L.-X., and Feng, Z.-H. (2010), "Dimension reduction in regressions through cumulative slicing estimation," *Journal of the American Statistical Association*, 105, 1455–1466.

Zhu, X., Huang, D., Pan, R., and Wang, H. (2020), "Multivariate spatial autoregressive model for large scale social networks," *Journal of Econometrics*, 215, 591–606.

Zhu, X., Pan, R., Li, G., Liu, Y., and Wang, H. (2017), "Network vector autoregression," *The Annals of Statistics*, 45, 1096–1123.

Zhu, X., Wang, W., Wang, H., and Härdle, W. K. (2019), "Network quantile autoregression," *Journal of econometrics*, 212, 345–358.

Zou, T., Lan, W., Wang, H., and Tsai, C.-L. (2017), "Covariance Regression Analysis," *Journal of the American Statistical Association*, 112, 266–281.

Table 3: The simulation results for the stochastic block network and the power-law distributed network. For each combination of $(N, n_{\max})$, we report the IMSE values ($\times 10^{-3}$) for $\rho(t)$, $\beta_1(t)$, $\beta_2(t)$, $\phi_1(t)$, $\phi_2(t)$ and $C(s,t)$, together with the MSE for λ_1 and λ_2 in the first line. We report the corresponding SUP values in the second line. The columns of $\rho(t)$, $\beta_1(t)$, $\beta_2(t)$, $\phi_1(t)$, $\phi_2(t)$ are original results multiplied by 10^{-3}.

Stochastic Block Model									
N	$n_{\max}$	$\rho(t)$	$\beta_1(t)$	$\beta_2(t)$	$C(s,t)$	λ_1	λ_2	$\phi_1(t)$	$\phi_2(t)$
100	50	0.220	0.049	0.136	2.531	0.234	0.246	3.326	3.793
		0.959	0.143	0.374	0.246	–	–	7.929	7.562
200	100	0.040	0.085	0.098	1.668	0.118	0.115	0.921	0.645
		0.133	0.304	0.219	0.157	–	–	2.177	1.509
500	200	0.016	0.082	0.085	1.232	0.060	0.059	0.033	0.034
		0.039	0.294	0.208	0.115	–	–	0.081	0.097
Power-law distributed Network									
N	$n_{\max}$	$\rho(t)$	$\beta_1(t)$	$\beta_2(t)$	C(s,t)	λ_1	λ_2	$\phi_1(t)$	$\phi_2(t)$
100	50	1.312	0.083	0.209	2.195	0.183	0.206	2.889	2.214
		4.936	0.169	0.479	0.223	–	–	6.719	4.857
200	100	0.168	0.015	0.055	1.420	0.096	0.088	0.339	0.213
		0.875	0.037	0.187	0.139	–	–	0.774	0.358
500	200	0.026	0.001	0.034	0.932	0.044	0.037	0.002	0.005
		0.101	0.001	0.079	0.087	–	–	0.007	0.014

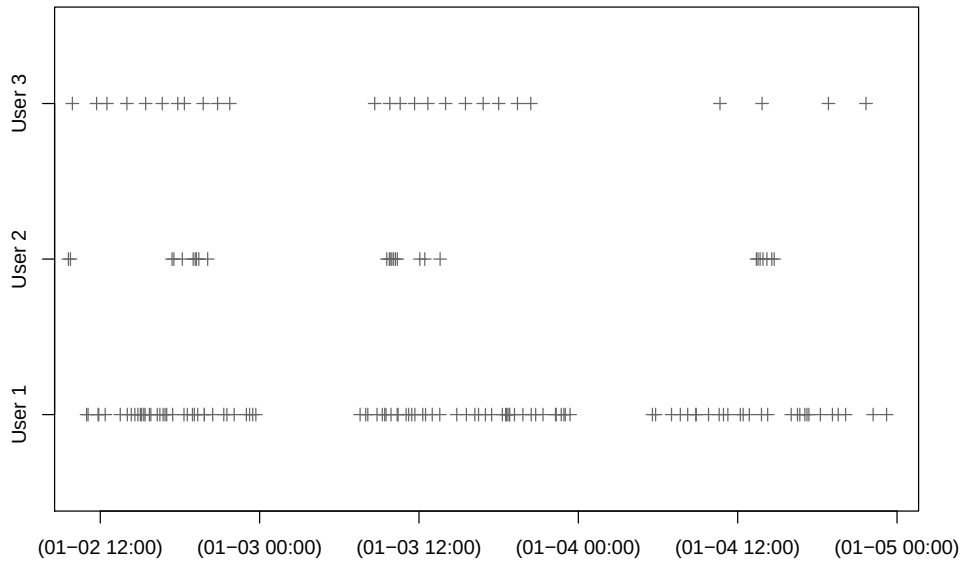

Figure 1: The posting time of three users from the data set.

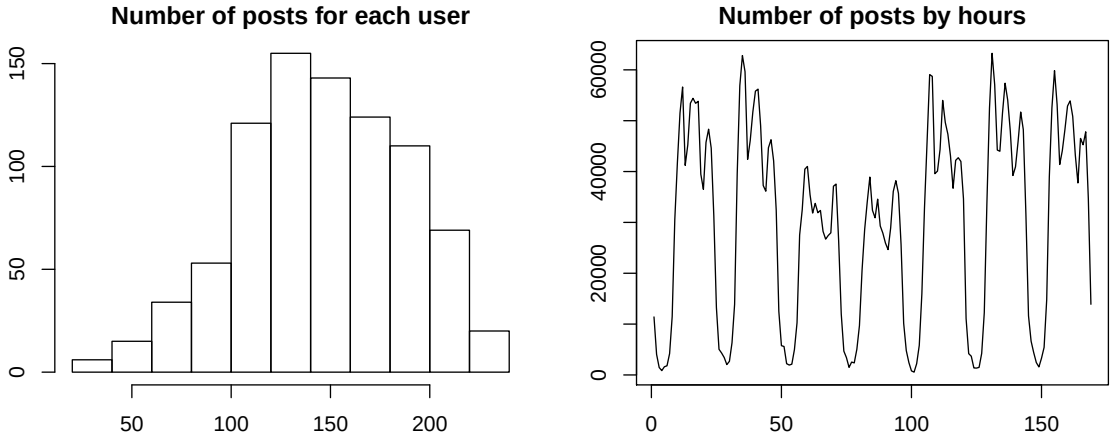

Figure 2: Left panel: total number of posts by each user; right panel: hourly aggregated number of posts as a function of time in hours.

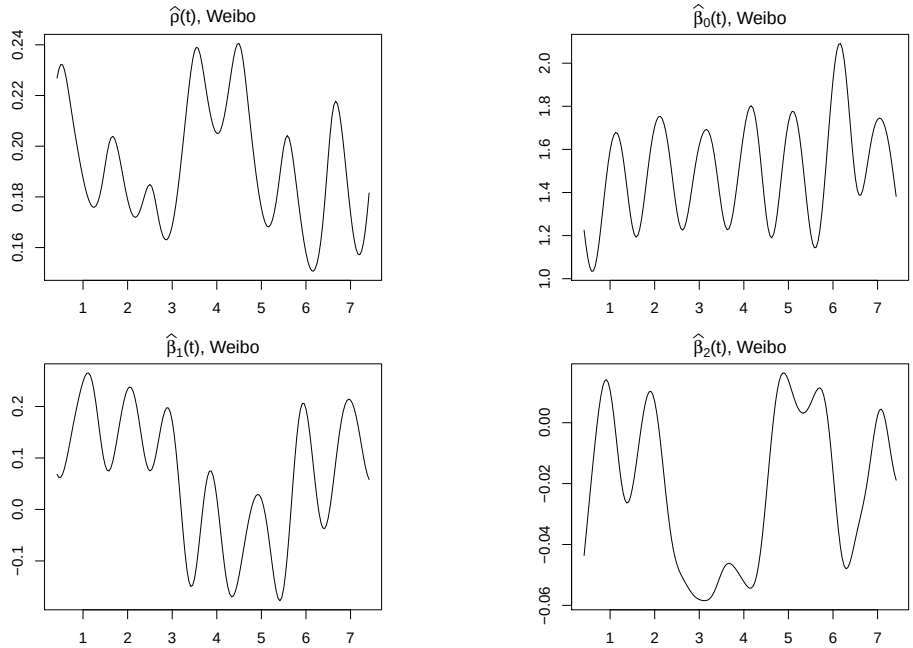

Figure 3: Estimated parameter functions $\widehat{\rho}(t)$, $\widehat{\beta}_0(t)$ (intercept), $\widehat{\beta}_1(t)$ (gender), $\widehat{\beta}_2(t)$ (number of labels) using the Weibo data.

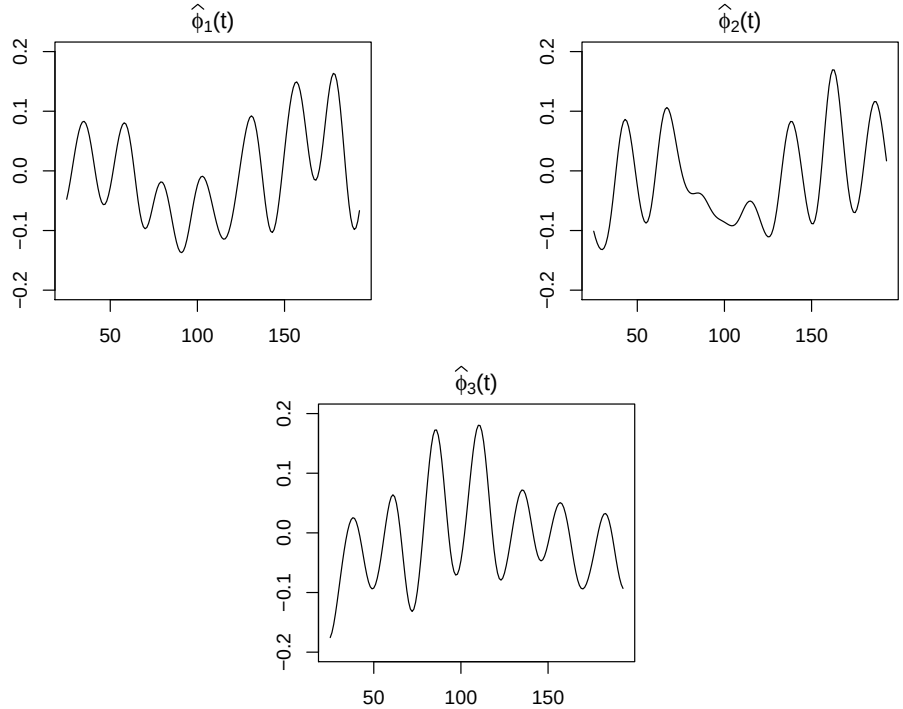

Figure 4: Estimates of the first three eigenfunctions for the Weibo data.